

Progress

AI-RAN GPU Isolation — Mechanism Deep-Dive

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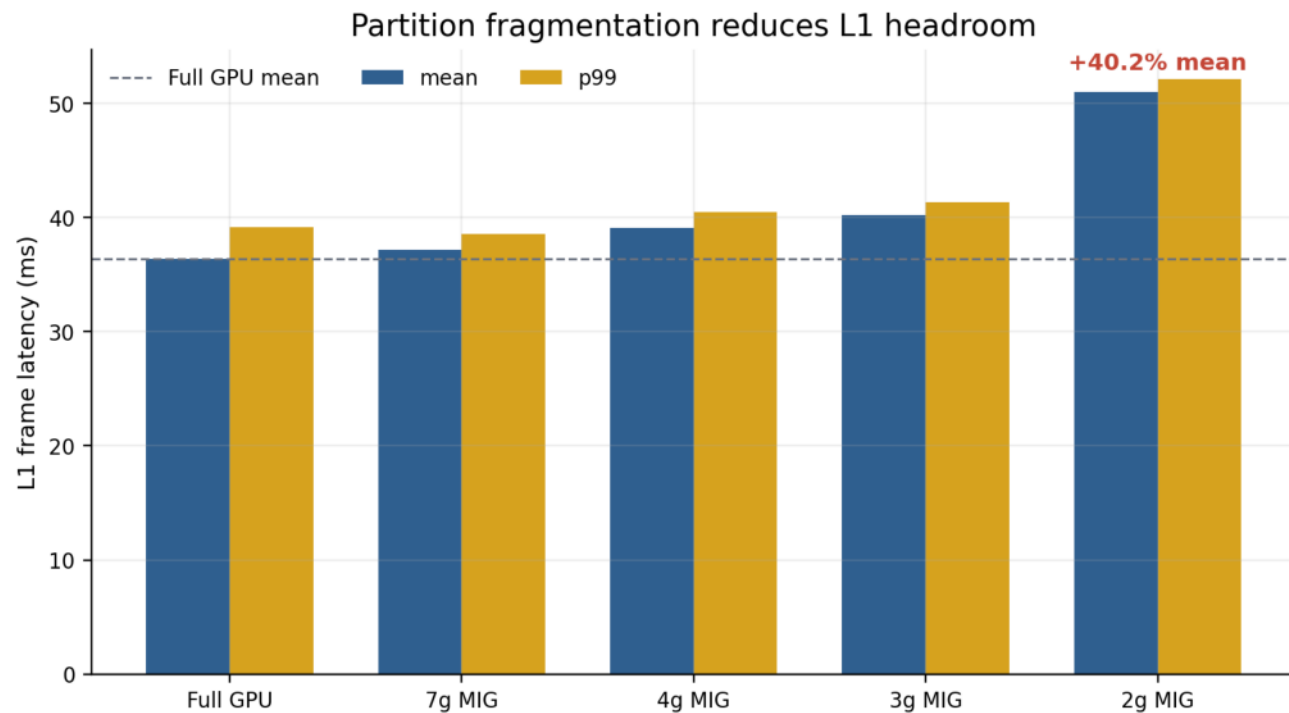
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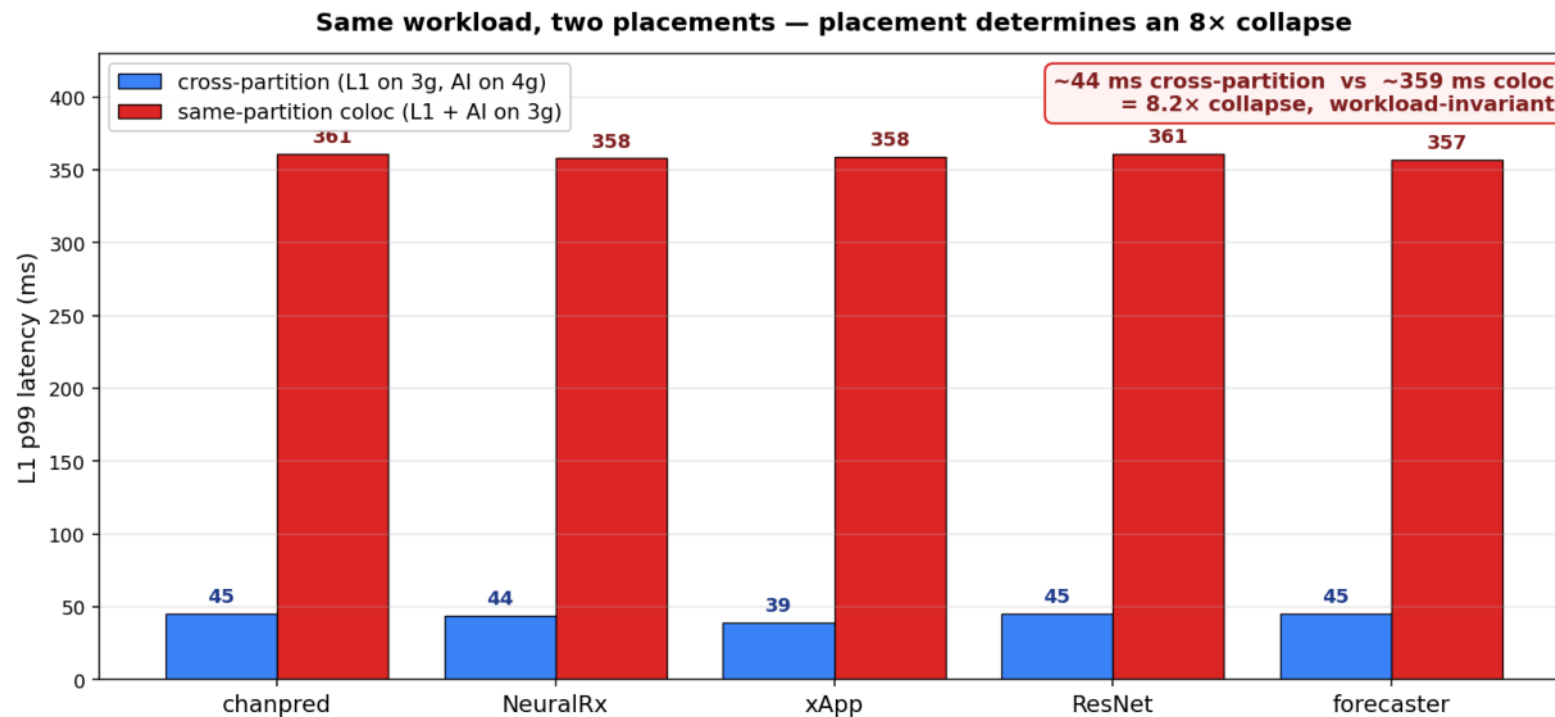
Today

- Continuation of Exp1-6 (5/29): MIG single-tenant overhead measured at +13-32 ms.
- Goal: Identify the hardware resource responsible for the L1 latency cost under AI co-tenancy.
- Setup: n=500-1000 per condition (25× the prior n=20), CloudLab d8545 (driver 550) + Perlmutter cross-validation.



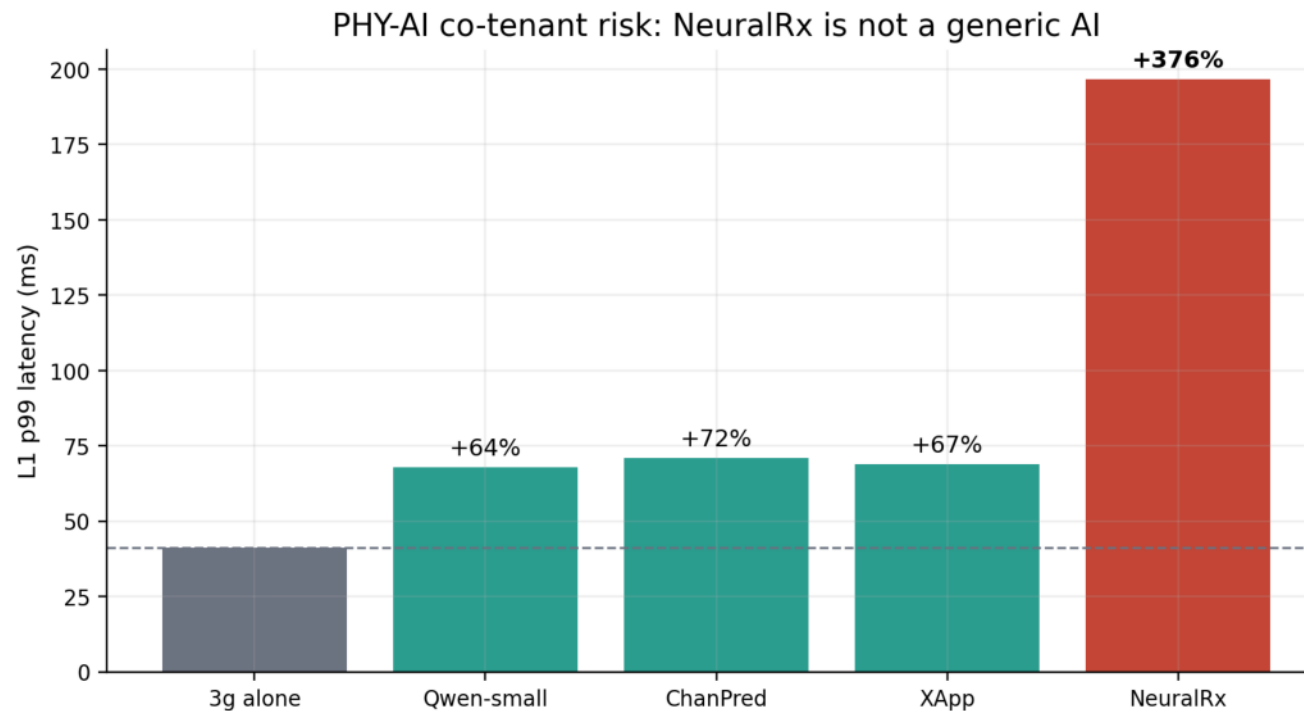
Placement Determines an 8× Collapse

- **Cross-Partition is Stable:** Five distinct AI workloads (chanpred, NeuralRx, xApp, ResNet, forecaster) all keep L1 p99 at ~45 ms when placed on a separate MIG partition.
- **Same-Partition Collapses 8×:** Identical workloads co-located in the same partition as L1 jump to ~358 ms.
- **The Cost Driver is Placement, Not Workload:** The contrast holds across all five workloads. The cost is set by where AI is placed.



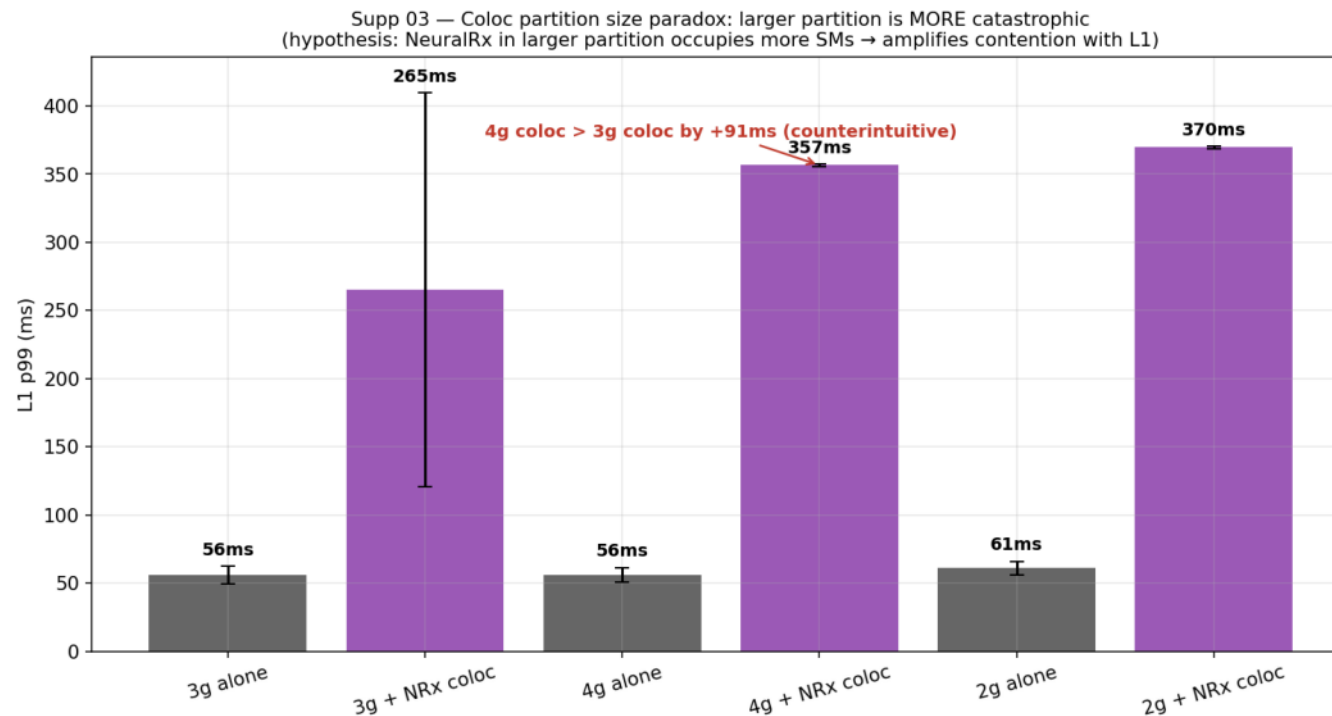
NeuralRx is Uniquely Dangerous Even Cross-Partition

- **PHY-AI Breaks Cross-Partition Isolation:** NeuralRx in a separate partition inflates L1 p99 by +376% (41 → 197 ms). ChanPred, xApp, and Qwen on the same setup only inflate L1 by +60-72%.
- **Why NeuralRx Specifically:** NeuralRx (TensorRT inline NN) shares the cuPHY-style copy/convert traffic pattern with L1, producing contention even across MIG partition boundaries.
- **Implication:** The MIG cross-partition isolation that holds for generic AI does not hold for PHY-AI co-tenants.



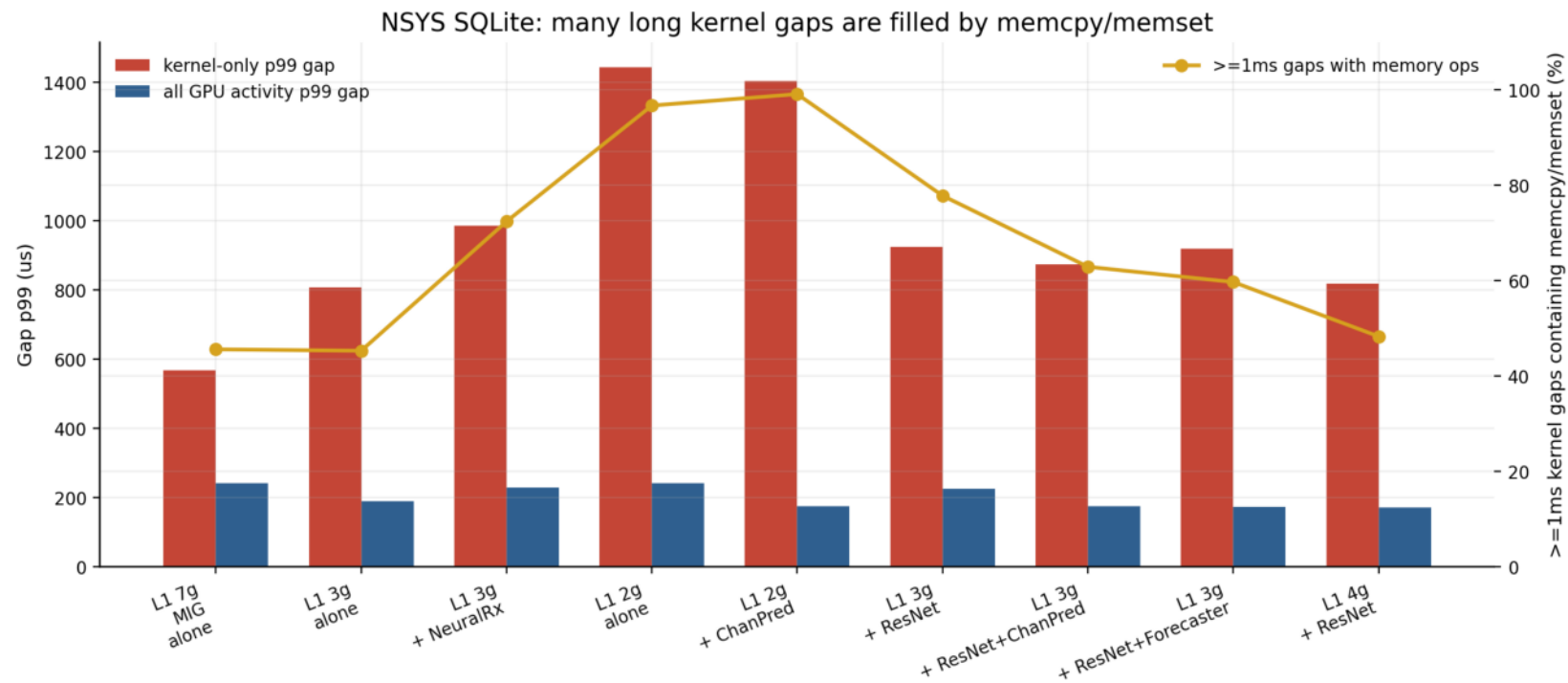
Larger Partition = MORE Catastrophic

- **Counter-Intuitive Outcome:** 4g co-location (371 ms) is worse than 2g co-location (369 ms). Adding SMs and HBM does not reduce the collapse.
- **Resource Allocation is Not the Lever:** Partition sizing controls the structural HBM share but not the contention path.
- **Bottleneck is Chip-Global:** The contended resource lies outside the per-partition abstraction MIG provides.



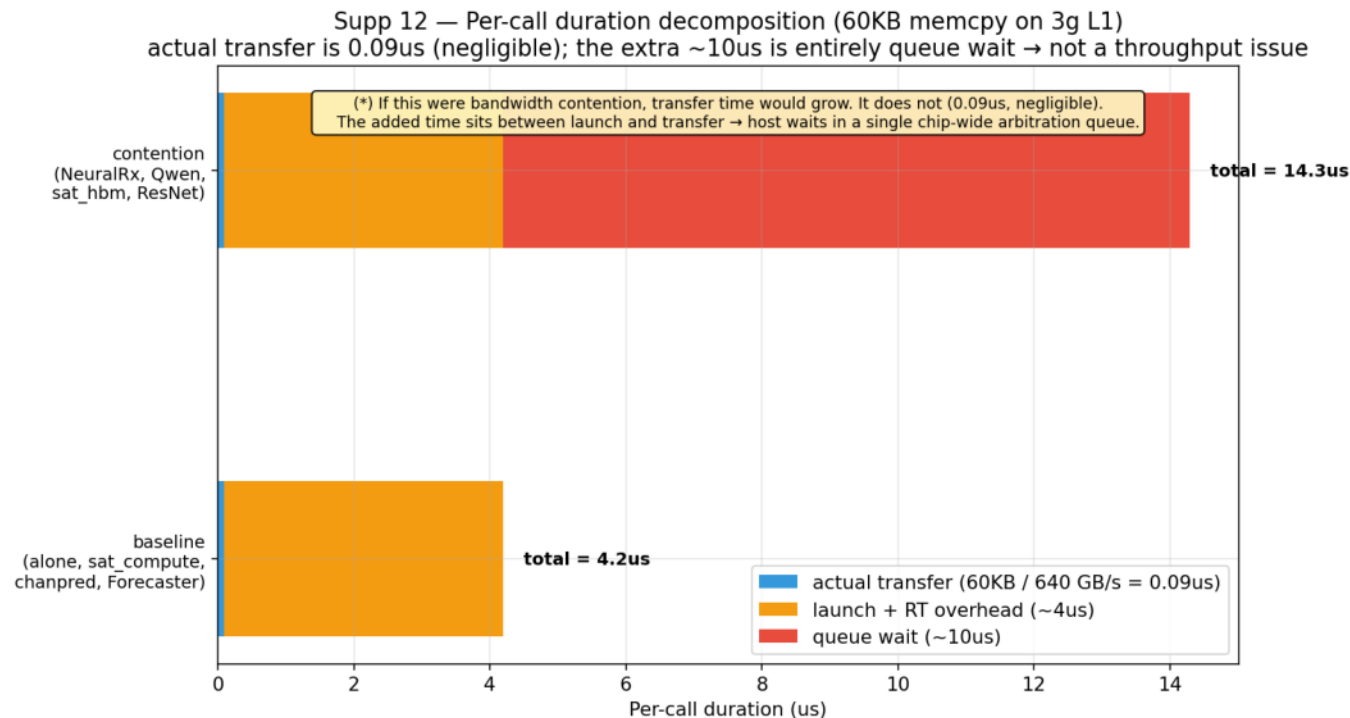
Mechanism — Gaps Grow at Memory-Op Boundaries with Same Op Count

- **Kernel Duration Unchanged:** Per-kernel execution time is identical under alone and co-location; only inter-kernel gaps grow.
- **Gaps Are Not Idle:** Gaps coincide with memcpy and memset boundaries — filled with memory ops, not idle time (confirmed across 30 NSYS captures).
- **Op Count is Invariant, Duration Inflates:** memcpy and memset call counts are identical (5,778 each); memcpy average duration rises from 4.2 μ s to 9.8 μ s (+133%). Same work, longer wait per operation.



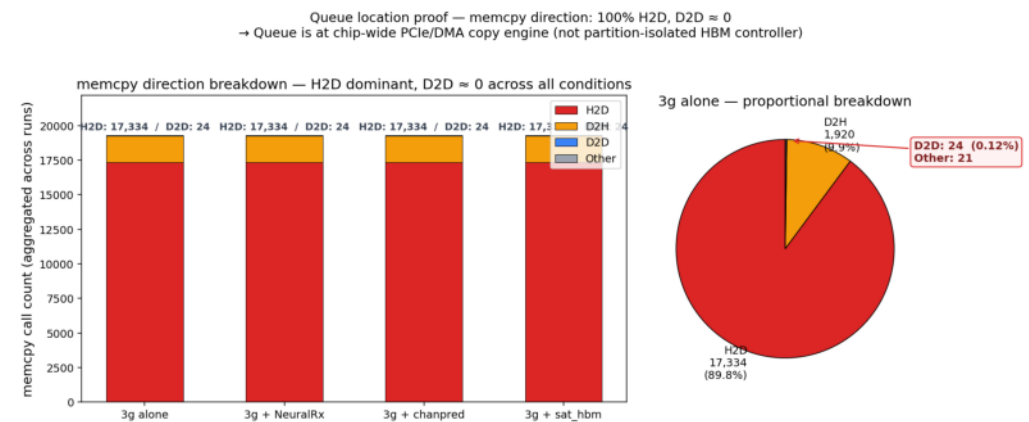
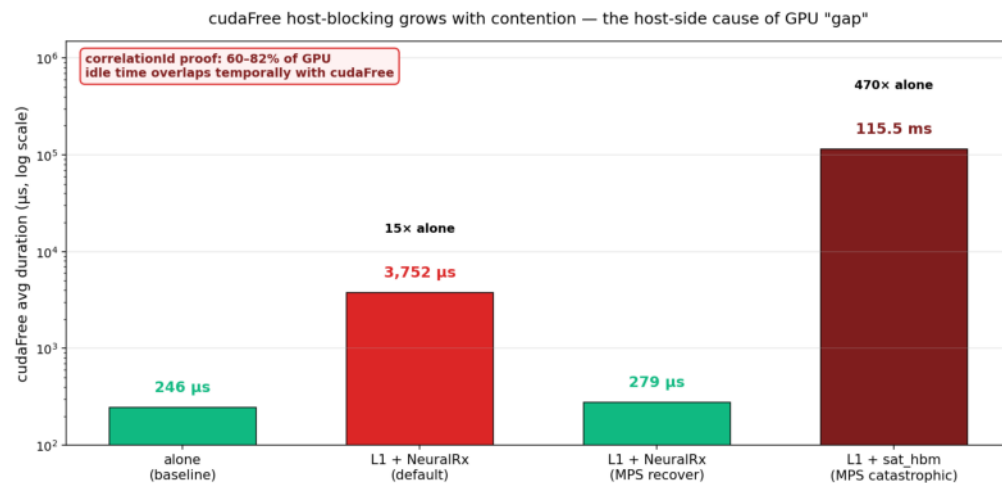
Mechanism — Arithmetic Proves Queue Wait, Not Throughput

- **Actual Transfer is Negligible:** $60 \text{ KB} / 640 \text{ GB/s} = 0.09 \mu\text{s}$. Bandwidth saturation cannot account for the inflation.
- **Baseline Decomposition:** $4.2 \mu\text{s}$ alone = launch + runtime overhead; transfer contributes essentially nothing.
- **Contended Decomposition:** $14.3 \mu\text{s} = 4.2 \mu\text{s}$ baseline + $\sim 10 \mu\text{s}$ queue wait. The slow mode is queue wait by construction.



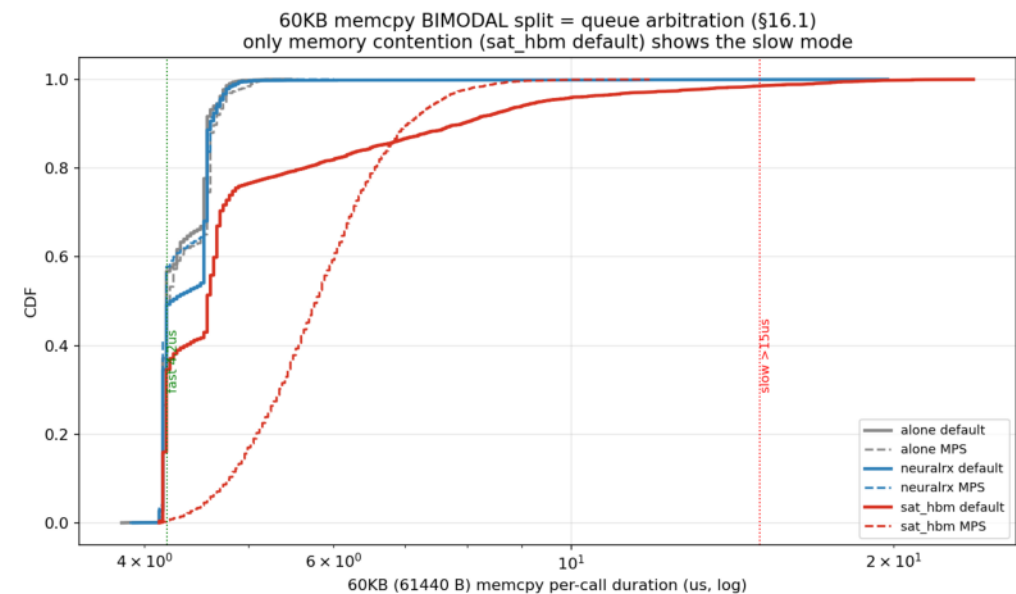
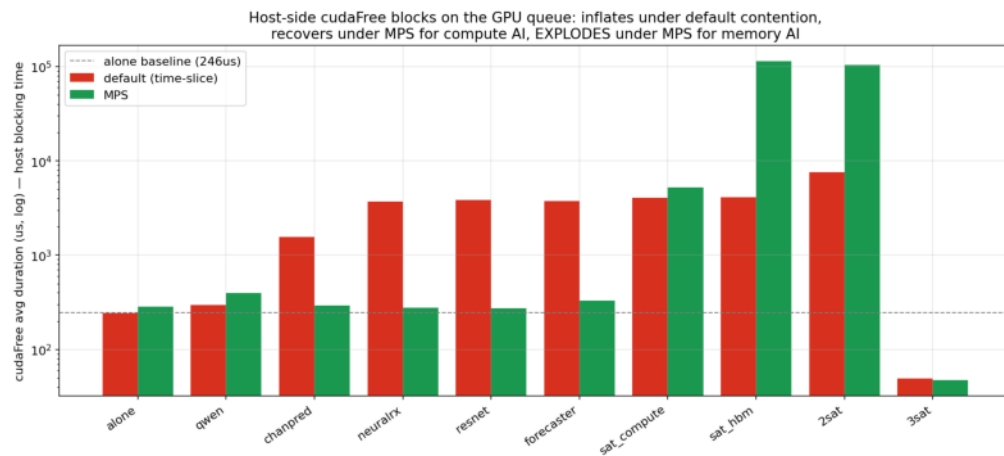
Mechanism — Queue at Chip-Wide PCIe/DMA + cudaFree Manifestation

- **Queue Location:** memcpy direction breakdown is 89.8% H2D, 9.9% D2H, 0.12% D2D. D2D ≈ 0 rules out the HBM controller; the queue is at the chip-wide PCIe/DMA copy engine.
- **Host-Side Manifestation:** cudaFree average grows 246 μ s (alone) $\rightarrow$ 3,752 μ s (NeuralRx, 15 $\times$) $\rightarrow$ 115,506 μ s (sat_hbm MPS, 470 $\times$). The same call recovers to 279 μ s under MPS + NeuralRx.
- **Causal Proof via correlationId:** 60-82% of GPU idle gap time aligns temporally with cudaFree blocking — the host-side block is the direct cause of the GPU-side gap.



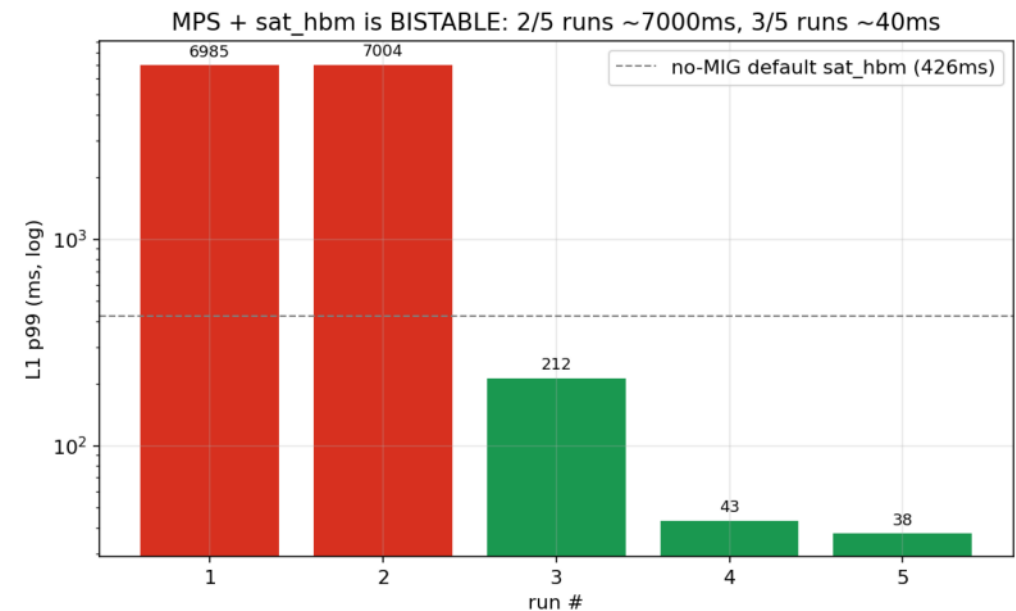
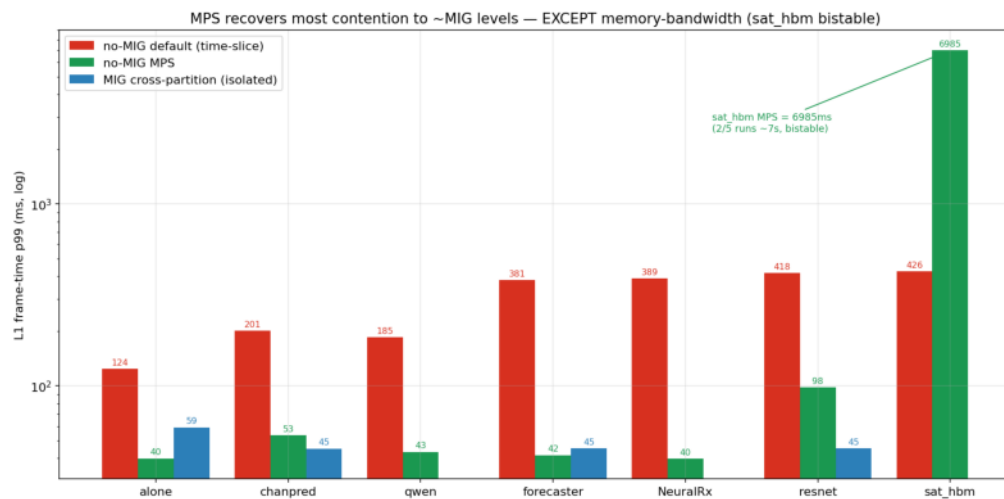
Cross-Platform Mechanism Validation

- **Bimodal Signature Reproduced:** Perlmutter no-MIG (different cluster, different driver) shows the same 60 KB memcpy bimodal split ($4.2 \leftrightarrow 16.8 \mu\text{s}$).
- **cudaFree Scaling Matches:** cudaFree average grows 15× under NeuralRx default and 469× under MPS + sat_hbm, mirroring CloudLab.
- **Conclusion:** The mechanism is NVIDIA-stack-wide. It is not specific to MIG or to the CloudLab driver.



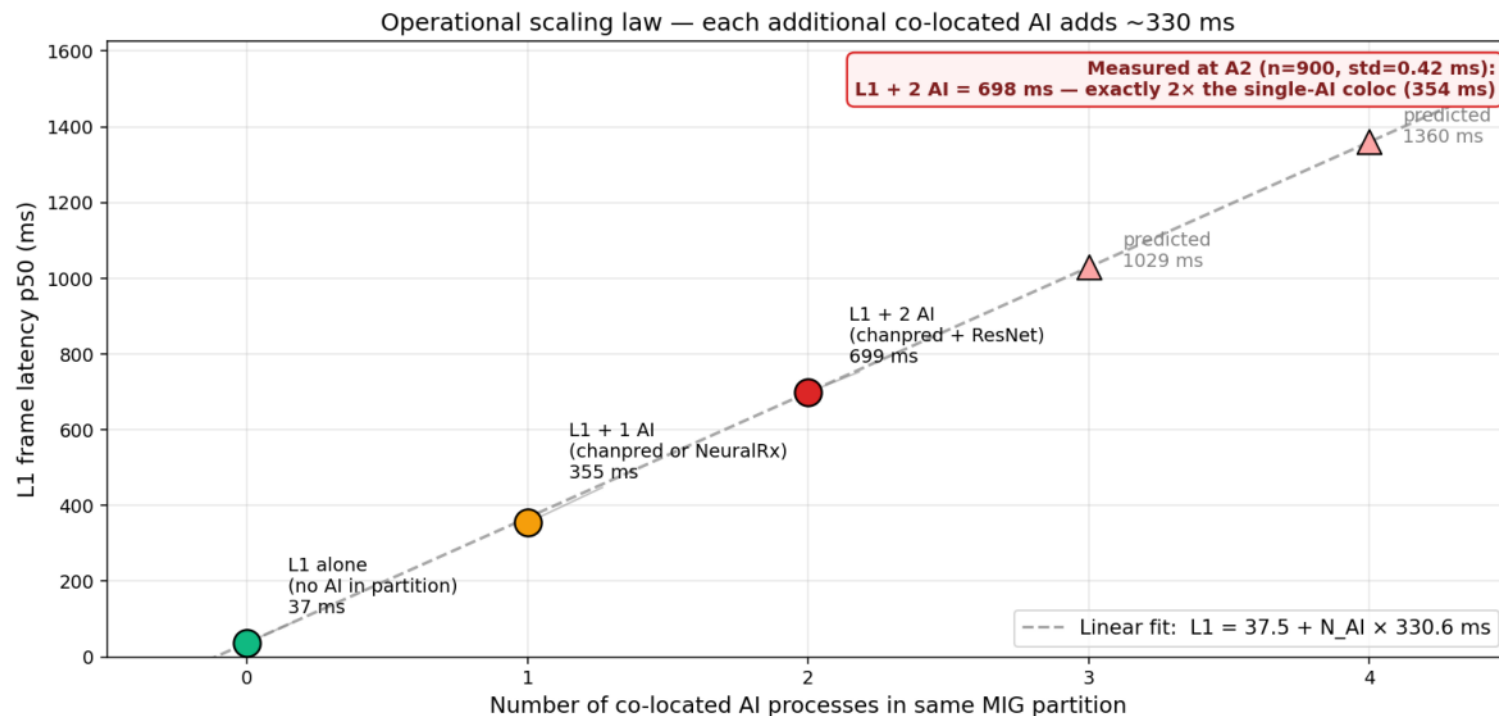
MPS Recovers Compute AI, Catastrophic for Memory AI

- **Compute AI Recovers:** NeuralRx 389 → 40 ms under MPS (below MIG cross-partition 197 ms); forecaster 381 → 42 ms; Qwen 185 → 43 ms.
- **Memory AI Becomes Worse:** sat_hbm 426 → 6,985 ms under MPS (bistable across runs).
- **Same Mechanism, Different Trigger:** MPS routes through the same cudaFree path; memory-bound workloads starve the device and re-engage the failure mode.



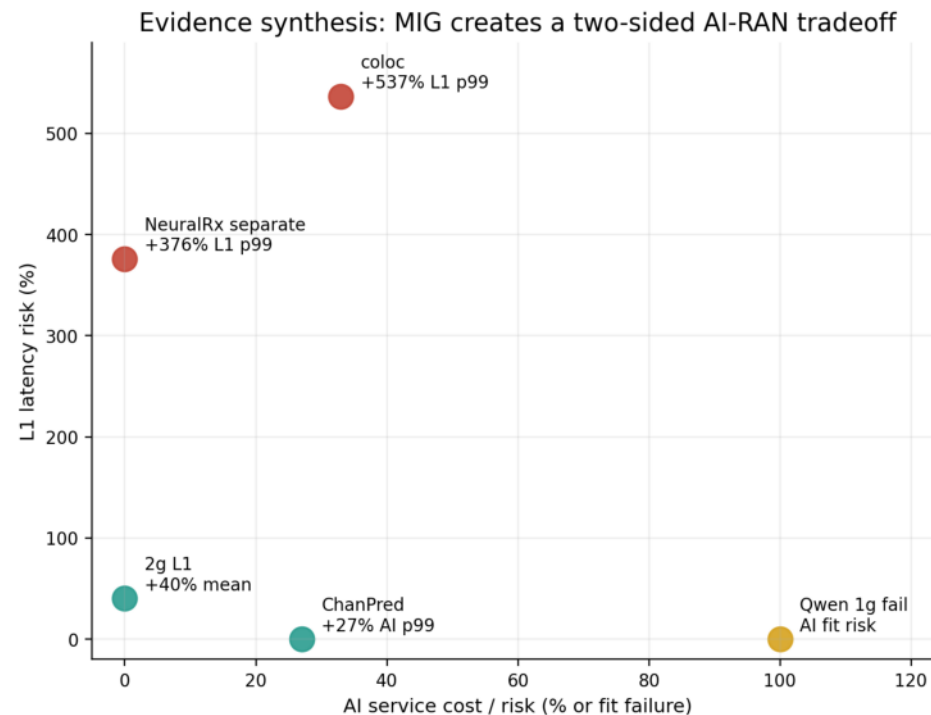
Operational Scaling Law — Each AI Adds ~330 ms

- **Linear Cost Model:** $L1 = 37.5 + N_AI \times 330.6$ ms when AI processes share the same MIG partition.
- **A2 Validates Slope:** L1 + 2 AI measures 698 ms (n=900, std=0.42 ms) — exactly 2× the single-AI co-location (354 ms).
- **Multi-Tenant Mistakes Compound:** Predicted 1,029 ms at 3 AI and 1,360 ms at 4 AI. Operational error scales linearly, not as a single step.



Cost Decomposition — Two Hardware-Level Costs

- **Structural Cost (memset):** Tied to partition HBM bandwidth share (2/7, 3/7, 4/7); selectable through partition sizing.
- **Contention Cost (memcpy):** Tied to the chip-wide PCIe/DMA copy engine; not isolable by any partition choice.
- **Unified Across Three Platforms:** MIG, no-MIG, and MPS each avoid one cost while remaining exposed to the other.



Design Rules

- **Placement Enforcement is Mandatory:** L1 and AI must reside in different MIG partitions; one misplacement produces the full 8× collapse.
- **Partition Sizing Serves Only L1 Budget:** Sizing has no mitigating effect under co-location; choose for L1 SM/HBM headroom only.
- **AI Workload Classification is Required:** Memory-bound AI cannot be safely co-located even under MPS.
- **MIG $\neq$ Chip-Wide Isolation:** MIG isolates per-partition SM/HBM but not the chip-global PCIe/DMA copy engine.

Thank You
